

EEE 359

Engineering Project Management

Lecture 06

Cost Estimation

Cost Estimating

Cost estimating is an essential part of project management, since it becomes the baseline for the subsequent cost control. If the estimate for a project is **too low**, a company may well **lose money** in the execution of the work. If the estimate is **too high**, the company may well **lose the contract** due to overpricing.

Estimating the cost of a project requires a **structured approach**, but whatever method is used, the first thing is to decide the **level of accuracy** required. This depends on the status of the project and the information available. There are **four main estimating methods** in use, varying from the very approximate to the very accurate. These are:

- **Subjective** (degree of accuracy $\pm 20\%$ – 40%)
- **Parametric** (degree of accuracy $\pm 10\%$ – 20%)
- **Comparative** (degree of accuracy $\pm 10\%$)
- **Analytical** (degree of accuracy $\pm 5\%$)

Subjective or Expert Judgement Estimating

At the proposal stage, the contractor may well be able to give only a '**ballpark figure**' to give a client or sponsor an approximate 'feel' of the possible costs. The estimating method used in this case would either be **subjective or approximate parametric**.

In either case, the **degree of accuracy** would largely **depend on the experience of the estimator** (the engineers managers). When using the subjective method, the estimator relies on his or her experience of similar projects to give a cost indication based largely on 'hunch'.

Geographical and political factors as well as the more obvious **labour and material** content must be taken into account. Such an approximate method of estimating is often given the disparaging name of '**guesstimating**'.

Parametric Estimating

The parametric method would be used at the **budget preparation stage**, but relies on good **historical data-based past jobs or experience**. By using well-known empirical **formulae or ratios**, in which costs can be related to specific characteristics of known sections or areas of the project, it is possible to produce a good estimate on which firm decisions can be based.

Clearly, such estimates need to be qualified to enable **external factors** to be separately assessed.

For example, an architect will be able to give a parametric estimate of a new house once he or she is given the volume ($\text{height} \times \text{length} \times \text{depth}$) of the proposed building and the standard of construction or finish. The estimate will be in £/cubic metre of structure. Similarly, office blocks are often estimated in £/square metre of floor space. The qualifications would be the location, ground conditions, costs of the land, etc.

Comparative or Analogous Estimating

As an alternative to the parametric method, the comparative method of estimating can be used for the preparation of the budget.

When a new project is very similar to another project recently completed, a quick comparison can be made of the salient features. This method is based on the costs of a **simplified schedule of major components** that were **used on previous similar jobs**. It may even be possible to use the costs of a similar-sized complete project of which one has had direct (and preferably recent) experience. Due allowance must clearly be made for the inevitable minor differences, inflation and other possible cost escalations.

An example of such a comparative estimate is the installation of a new computer system in a building when an almost identical (and proven) system was installed 6 months earlier in another building. It must be stressed that such an estimate does not require a detailed breakdown.

Analytical or Bottom-Up Estimation

Once the project has been **sanctioned**, a **working budget** estimate will be necessary against which the cost of the project will be controlled. This will normally require an **analytical estimate or bill of quantities**. This type of estimate may also be required where a contractor has to submit a fixed-price tender, since once the contract is signed there can be no price adjustment except by inflation factors or client-authorized variations.

As the name implies, this is the most accurate estimating method, but it requires the project to be broken down into **sections, subsections** and finally **individual components**. Each component must then be given a cost value (and preferably also a cost code) including both the material and labour contents. The values, which are sometimes referred to as 'norms', are usually extracted from a database, or company archives, and must be individually updated or factored to reflect the current political and environmental situation.

Three Point Estimating

The accuracy of single-point cost or duration estimates may be improved by considering estimation uncertainty and risk. Using three-point estimates helps define an approximate range for an activity's cost or duration:

- **Most likely (M):** Realistic estimate, assuming normal conditions.

This estimate is based on the duration of the activity, given the resources likely to be assigned, their productivity, realistic expectations of availability for the activity, dependencies on other participants, and interruptions

- **Optimistic (O):** Best case scenario, assuming everything goes perfectly.
- **Pessimistic (P):** Worst case scenario, assuming things go wrong.

One commonly used formula is:

$$E = (O + 4M + P) / 6$$

Configuration Management

Configuration Management

Describes how the information about the project components (hardware, software, documents) will be recorded and updated so that the product, service, or result of the project remains consistent, controlled, operative and traceable throughout the project lifecycle.

Configuration management reduces errors and rework, improves coordination among team members and between teams, and ensures project quality is maintained throughout.

Keeping track of all these changes to specifications, drawings, support documentation and manufacturing processes is the essence of configuration management, which can be split into the following **five main stages**:

- **Configuration management planning:** This covers the necessary standards, procedures, support facilities, resources and training, and sets out the scope, definitions, reviews, milestones and audit dates. Part of the Project Management Plan document.

Configuration Management

- **Configuration identification:** The items that will form part of the configuration library (plans, drawings, specifications etc) is identified and catalogued. This encompasses the logistics and systems and procedures. It also defines the criteria for selection in each of the project phases.
- **Configuration-change management:** This deals with the proposed changes and their review before acceptance re ection. nly approved changes are implemented. At this stage, changes are compared with the configuration baseline.
- **Configuration-status accounting:** This records and logs the current version and the accepted (registered) changes of each item, as well as providing traceability of all baselines.
- **Configuration audit:** This ensures that all the previous stages have been correctly applied and incorporated in the organization. The output of this stage is the audit report. Ensures that the entire configuration process is working as intended, the documents are maintained correctly and all members involved are aware of their roles.

Configuration Management

1. Management and Planning:

- Configuration Management plan document

2. Identification:

- Example: For a UAV project: components include drone body, flight control software, battery system, delivery mechanism etc.

3. Control:

- Change Request forms, Change Control Board (CCB)

4. Status Accounting:

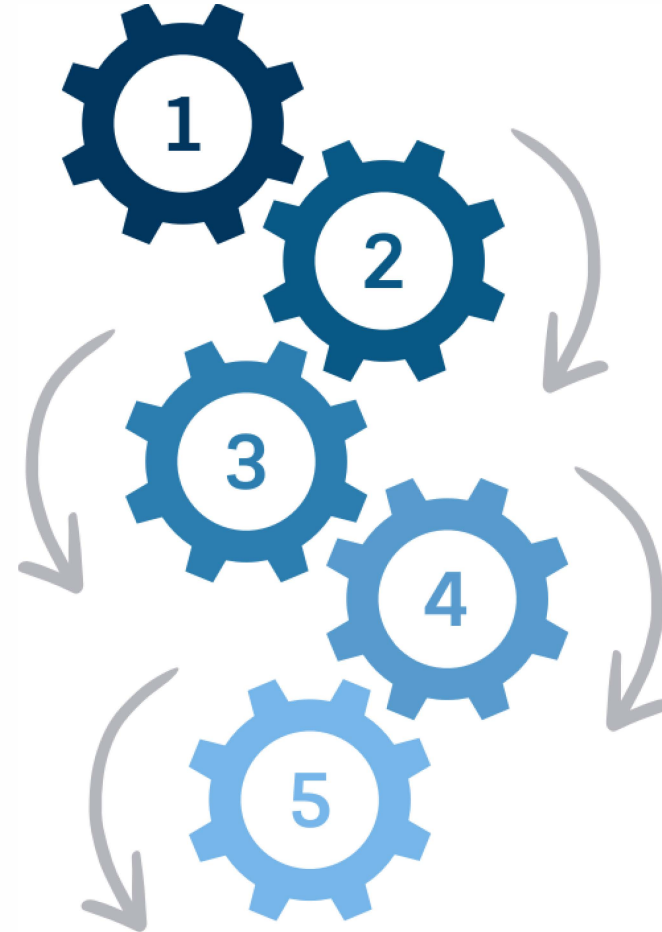
- software: v1.0 -> v1.1 -> v2.0. Record who changed what and when.

5. Audit:

- Functional (product works or not) and physical (product built correctly or not).

Configuration Management Process

- Plan**
Document Config Management approach
- Identify**
Identify configuration record items
- Control**
Manage access and maintain audit trail
- Status Accounting**
Log status of records at all times
- Audit**
Check records meet original requirements



Thanks